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AgriSphere – Smart Agricultural Collaboration Platform

CSA0904-Programming in Java

Supervised By

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SIMATS Engineering

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 GPS Map Camera



Mevalurkuppam, Tamil Nadu, India



22g8+cwh, Mevalurkuppam, Tamil Nadu 602105, India

Lat 13.026456° Long 80.016866°

Wednesday, 05/08/2026 03:00 PM GMT +05:30



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SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

AgriSphere AI – Intelligent Digital Agriculture Ecosystem

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Introduction



- AgriSphere AI is an intelligent digital agriculture ecosystem that connects farmers, buyers, and service providers on a single platform.
- The platform enables farmers to create digital profiles and manage their agricultural products efficiently.
- Buyers can easily search, compare, and purchase agricultural products from trusted farmers.
- AI technology provides personalized recommendations based on user preferences and market trends.
- Smart notifications keep users informed about orders, payments, and important agricultural updates.
- Analytics dashboards help users monitor business performance and make informed decisions.
- The project promotes digital transformation by making agriculture smarter, faster, and more efficient.



PROBLEM STATEMENT



- Farmers often struggle to find genuine buyers and receive fair prices for their products.
- Buyers face difficulty in identifying trusted farmers and quality agricultural products.
- Agricultural services such as equipment rental and consultancy are not available on a single platform.
- Traditional farming and selling methods reduce efficiency and increase operational costs.
- Farmers have limited access to AI-based recommendations and real-time market insights.
- Existing agriculture applications do not provide complete digital solutions for all stakeholders.
- A centralized digital ecosystem is required to improve communication, sales, and agricultural productivity.



Project Scope & Objectives

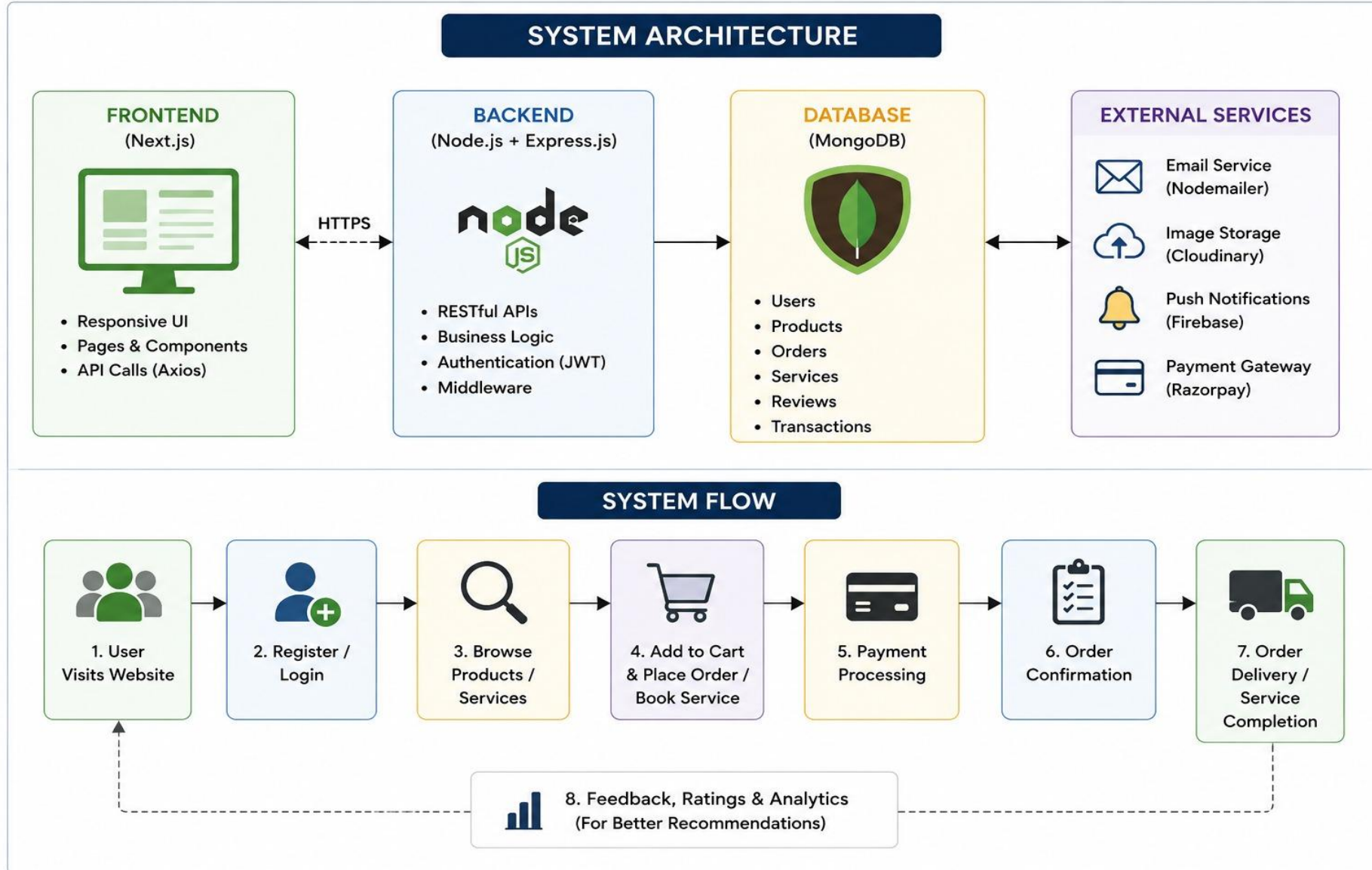


- Develop a centralized platform that connects farmers, buyers, service providers, and administrators.
- Allow farmers to manage products, receive orders, and track their sales digitally.
- Provide buyers with a secure marketplace to purchase agricultural products.
- Integrate AI to deliver personalized recommendations and market insights.
- Implement smart notifications for orders, payments, and important updates.
- Generate analytics reports to help users monitor performance and business growth.
- Build a scalable system that can support future technologies like IoT and weather forecasting.

LITERATURE REVIEW

S. No	LITERATURE TITLE	AUTHOR AND YEAR	LIMITATIONS	HOW OUR PROJECT OVERCOMES IT
1	Smart Agriculture Monitoring Using IoT and Cloud Computing	Patel & Singh, 2022	Focuses only on farm monitoring and sensor data without providing an online marketplace.	Integrates smart farming with an AI-powered marketplace connecting farmers, buyers, and service providers.
2	AI-Based Crop Recommendation and Yield Prediction System	Kumar et al., 2023	Provides crop recommendations but does not support product selling or business management.	Combines AI recommendations with product management, buyer discovery, and business analytics.
3	Online Agricultural Marketplace for Farmers	Sharma & Verma, 2021	Supports online product selling but lacks AI recommendations and service provider integration.	Offers AI-based recommendations, service booking, smart notifications, and a complete digital agriculture ecosystem.
4	Digital Farming Platform with Precision Agriculture	Das & Roy, 2024	Concentrates on precision farming techniques without connecting buyers and agricultural services.	Connects farmers, buyers, and service providers while providing analytics, AI insights, and an integrated marketplace.

System Architecture





MODULES



Module 1 – User & Profile Management

Provides secure registration, login, and role-based access for farmers, buyers, service providers, and administrators. Users can create and manage their profiles within the platform.

Module 2 – Smart Digital Marketplace

Enables farmers to list agricultural products while buyers can search, compare, and purchase them through an easy-to-use online marketplace.

Module 3 – AI Recommendation Engine

Uses user preferences and browsing history to provide personalized product recommendations, improving product discovery and customer experience.

Module 4 – Order & Service Management

Handles shopping cart operations, order placement, payment processing, service bookings, and real-time order tracking for users.

Module 5 – Analytics & Administration

Provides administrators with dashboards to monitor users, products, orders, and sales while generating reports and business insights for better decision-making.



Evaluation Metrics



- Measure the success rate of user registration and secure login functionality.
- Evaluate the performance of product listing, searching, and order management features.
- Analyze the accuracy and usefulness of AI-generated recommendations.
- Measure order processing time and successful transaction completion.
- Collect user feedback to evaluate satisfaction and platform usability.
- Assess the performance and accuracy of the analytics dashboard.
- Ensure the system provides reliable, secure, and efficient services to all users.



Expected Outcomes



- Farmers can directly connect with buyers and increase their income through digital sales.
- Buyers can easily access quality agricultural products from trusted farmers.
- Service providers can efficiently offer their agricultural services through the platform.
- AI recommendations will help users make better farming and business decisions.
- Smart notifications will improve communication and reduce delays in transactions.
- Analytics dashboards will provide valuable insights into sales and business performance.
- The platform will encourage smart, sustainable, and technology-driven agriculture.



Conclusion



- AgriSphere AI is a complete digital agriculture ecosystem.
- It connects farmers, buyers, and service providers in one platform.
- AI improves efficiency through smart recommendations and analytics.
- The marketplace simplifies agricultural buying and selling.
- Multiple user roles make the system flexible and scalable.
- The project supports future expansion with advanced AI features.
- AgriSphere AI contributes to modern, smart, and sustainable agriculture.

Thank You!